

5 Results and Analysis

5.1 Experiments and Results

5.1.1 Experimental Design

We built a system to check if removing one power wire will break the network (N-1 contingency screening). To find out the true answer (label), the system must run a full math simulation (power-flow solve) for every single wire. We tested this on three different network maps (grid topologies) to see how well the artificial intelligence adapts to new places:

- **Training Map:** 36 stations (buses) and 59 wires (lines) (NeurIPS 2020), which gave us 702,618 test cases (contingency labels).
- **Small Unseen Map:** 14 stations and 20 wires (case14).
- **Large Unseen Map:** 118 stations and 186 wires (WCCI 2022).

To build the safety layer, our text-reading tool (LLM extraction pipeline) read 16 official grid documents. This produced 2,463 potential rules (candidates), which were heavily filtered down to 4 rules that actually worked with our simulator.

5.1.2 Models and Baselines

Our main artificial intelligence (Graph Neural Network or GATv2) uses 3 layers. It does not squish the map into one summary (no pooling) because almost all moments in time (96-99% of frames) have a mix of safe and unsafe wires (strictly mixed frames). Instead, it makes a separate guess for every single wire (edge-level readout).

We compared our artificial intelligence against two simple guesses (baselines):

- Guessing that everything will break (all-positive baseline).
-
- Using the single best math rule about the removed wire (best single rule).
-

[PLACEHOLDER: EXTERNAL STATE-OF-THE-ART BASELINES. Example text to add later: "We also evaluated our framework against external state-of-the-art models such as [Model A] and [Model B] to demonstrate

that our baseline neural network matches or exceeds current industry standards for this specific screening task."]

5.1.3 Metrics and Evaluation Regimes

We measured the artificial intelligence using a fairness score (F1 score). To keep the test completely fair, we locked the passing grade (held threshold) at 0.8849 based only on the training map, and used it unchanged everywhere. We avoided picking the best boundary for each individual map (oracle threshold) because that requires looking at the answer key.

We also locked the number of examples the artificial intelligence looks at together (eval batch size) to exactly 64. Changing this size changes the artificial intelligence's internal math (live batch statistics) and artificially shifts the final scores.

The safety gate is evaluated using three specific metrics:

- **Intervention Precision:** When the safety gate overrides the model, how often its decision is correct.
- **Reach:** The percentage of model predictions where the safety gate is eligible to take action.
- **Structural Ceiling:** The percentage of the model's missed violations that happen on a grid state that a rule can actually see.

5.1.4 Cross-Topology Results

The artificial intelligence performs well at home but struggles in new places (cross-topology degradation).

Topology	Lines	All-Positive Baseline	Best Single Rule	Model F1 Score
NeurIPS 2020 (Trained On)	59	0.3104	0.4639	0.8956

case14 (Unseen, Smaller)	20	0.4345	0.5392	0.4167
WCCI 2022 (Unseen, Larger)	186	0.3969	0.4915	0.5577

On the small unseen map (case14), it fails completely. It actually loses to the simple rule and to guessing that everything will break (all-positive baseline). Moving to a bigger map (scaling up) hurts the artificial intelligence less than moving to a smaller map (scaling down).

[PLACEHOLDER: STANDARD DEVIATIONS ACROSS MULTIPLE SEEDS.

Example text to add later: "To verify the stability of the neural network, training was repeated across 3 random seeds. The model achieved a score of 0.8956 ± 0.005 , confirming that the high performance on the training grid is a stable property of the architecture rather than a product of lucky initialization."

5.1.5 Symbolic Shield Performance

The safety gate only steps in to change a "safe" guess to a "dangerous" guess (asymmetric gate). It never blocks a warning.

Topology	Model F1	Shielded F1	Delta	Missed Violations	Intervention Precision	Reach
NeurIPS 2020	0.8956	0.9038	+0.0082	2,041 to 1,710	93.8%	0.31%
case14	0.4167	0.4188	+0.0021	18,592 to 18,497	92.2%	0.087%

WCCI 2022	0.557 7	0.6253	+0.067 6	68,166 to 47,101	93.4%	3.04%
--------------	------------	--------	-------------	---------------------	-------	-------

When we add the safety gate to the artificial intelligence, the overall score improves significantly on the large unseen map (WCCI 2022), successfully removing 31 percent of the dangerous mistakes. The gate's accuracy (intervention precision) stays completely flat. It is correct roughly 93 percent of the time regardless of the map. What changes is how often it needs to speak (reach). The artificial intelligence makes so many errors on the large map that the gate's reach jumps from 0.31 percent to 3.04 percent.

To prove this is a reliable physics rule and not just repeating the answer (label restatement), we measured that a broken map has a 91 to 96 percent chance of failing again, not 100 percent. This proves the rule carries real physical meaning. To further prove reliability, we ran the whole rule-finding process twice. Even though the second run found a different number of starting rules (29 percent difference), the final safety numbers were exactly identical down to the digit (bit-reproducible).

[PLACEHOLDER: ZERO-SHOT CAPABILITY RETENTION / EXTERNAL BENCHMARKS. Example text to add later: "To ensure the addition of the symbolic shield did not cause catastrophic forgetting or degrade the model's core processing abilities, we evaluated the shielded model against external diagnostic suites such as [Benchmark A]. The utility drift was minimal ($\pm 0.5\%$), proving the intervention is highly localized."]

5.1.6 Ablation Study

We turned off different parts of the system to prove they are necessary.

- **Communication Test (Message Passing Ablation):** When we stopped the network points from talking to each other, the score dropped from 0.8987 to 0.8411, proving the map structure carries vital information.
-
- **Rule Count Test:** We tested an uncleaned list of 32 rules versus a cleaned list of 4 rules. Throwing out 28 bad rules actually made the system better, proving that a rulebook full of bad rules completely hides the good ones.

-
- **Cost of Safety (Threshold Objective):** We checked what happens if we force the artificial intelligence to catch more problems (F2 score). We measured that catching one extra real problem costs 3.2 extra false alarms on the training map, showing exactly how much extra safety costs (F-beta trade curve).
-

5.1.7 Discussion

The safety gate solves the artificial intelligence's biggest weakness. While the artificial intelligence gets confused on new maps, the physical rules work perfectly everywhere (topology invariance).

We found a massive gap between the official rulebooks and our simulator (standards/simulator mismatch). We started with 2,463 rules but only kept 4. This is because 54 percent of the rules required measuring time or waves (frequency), which our simulator does not track (missing capabilities). Also, official rulebooks focus on how equipment connects (connection codes) rather than real-time safety (operational security).

We had to throw out rules about electrical pressure (voltage). Official rules say pressure must stay within a strict 5 percent window, but our simulator naturally runs 6 percent higher than the standard baseline (nominal voltage). The official rules caused massive false alarms on healthy maps. During testing, our text-checker (validator) rejected 31 out of 32 rules. We discovered the artificial intelligence was completely inventing fake reasons for rejecting them (fabricated reasoning), such as claiming a word was undefined even though it was given in the instructions. This proves we must always record the artificial intelligence's thinking.

Our final safety rule (thermal check) is highly trusted. We built a tracking map (knowledge graph) that proves this exact rule is written in 10 different sections across 4 official documents by 2 international rule-making groups.

Our system cannot catch every mistake. Between 69 and 99 percent of the artificial intelligence's dangerous errors happen on maps that look perfectly healthy before the wire breaks. Because our rules only look at the current picture (present state), they are completely blind to these future problems. Adding more rules will never fix this hard limit (structural ceiling).

Finally, we proved that trying to sort problems into 4 buckets (fault classification) was a bad idea because four simple math rules could solve it perfectly (closed-form target). We also proved that predicting future breaks (forecasting) is impossible because the simulator breaks wires randomly (unconditional Bernoulli draw).

● VERSION 2

5.1 Performance Evaluation

The artificial intelligence model [Graph Neural Network or GNN] was tested on three different power grid maps to measure how well it identifies dangerous broken wires [N-1 contingency screening]. On the familiar map it learned from [36-bus NeurIPS 2020], the model achieved a very strong fairness score [F1 score of 0.8956]. However, when tested on a much smaller, unfamiliar map [14-bus case14], the model failed completely, dropping to a score of 0.4167. On this small map, the model performed worse than simply guessing that every single wire break would cause a disaster [all-positive baseline]. On the much larger unfamiliar map [118-bus WCCI 2022], the model struggled but still functioned, scoring 0.5577.

Map [Topology]	Wires [Lines]	Guess Everything Breaks [All-Positive Baseline]	Basic Math Check [Best Single Rule]	AI Score [Model (Held) F1]
Familiar [NeurIPS 2020]	59	0.3104	0.4639	0.8956
Small Unfamiliar [case14]	20	0.4345	0.5392	0.4167

Large Unfamiliar [WCCI 2022]	186	0.3969	0.4915	0.5577
------------------------------------	-----	--------	--------	--------

The safety check [symbolic shield] performed incredibly well across all environments. When the safety gate stepped in to change a model's prediction, it was correct 93.8 percent of the time on the familiar map, 92.2 percent of the time on the small unfamiliar map, and 93.4 percent of the time on the large unfamiliar map [intervention precision]. On the largest unfamiliar map, the safety check successfully reduced the number of dangerous mistakes [unsafe-permissive predictions] by 31 percent.

Map [Topology]	AI Score [Model F1]	Protect ed Score [+Shield F1]	Improvem ent [Delta]	Missed Disaste rs [Missed Violatio ns]	Accuracy When Acting [Intervent ion Precision]	How Often It Acts [Reac h]
Familiar [NeurIPS 2020]	0.8956	0.9038	+0.0082	2,041 to 1,710	93.8%	0.31 %
Small Unfamili ar [case14]	0.4167	0.4188	+0.0021	18,592 to 18,497	92.2%	0.087 %
Large Unfamili ar	0.5577	0.6253	+0.0676	68,166 to 47,101	93.4%	3.04 %

[WCCI 2022]						
----------------	--	--	--	--	--	--

To prove this result is a solid rule of physics and not a lucky accident, the entire rule-gathering process was run a second time. Even though the second run found a completely different number of starting rules (a 29 percent difference), the final safety gate made the exact same number of corrections and false alarms down to the single digit [bit-reproducible].

5.2 Analysis of Design Solutions

Our first design attempted to sort grid problems into four separate buckets [4-class fault classification]. We threw this approach away because the answers could be perfectly calculated using four simple math rules [closed-form function] with zero disagreements across 315,000 records. Because the math solved it perfectly, the artificial intelligence was not actually doing any real work. We also tried predicting if a wire would break in the future [forecasting], but this was impossible to learn because the simulator breaks wires entirely by random chance [unconditional Bernoulli draw]. We ultimately chose a task that asks if removing a specific wire will break the grid right now [N-1 contingency screening].

To build the safety rules, our text-reading tool [LLM extraction pipeline] gathered 2,463 potential rules [candidates] from 16 official grid documents. After strict filtering, this massive list was reduced to 4 distinct rules that actually worked with our simulator. We found a massive gap between official rulebooks and our simulator [standards/simulator mismatch]. We had to throw out 1,332 rules simply because they relied on time and wave measurements [frequency], which our simulator does not track [missing capabilities].

During this phase, we found a major hidden problem with rules about electrical pressure [voltage]. Official guidelines say electrical pressure must stay within a strict plus or minus 5 percent window. However, simulated grids naturally operate about 6 percent higher than the standard baseline [nominal voltage]. Applying the official textbook rules directly to the simulator caused massive false alarms on perfectly healthy grids [base-kV contract].

[PLACEHOLDER: COMPARISON AGAINST EXTERNAL SOTA BASELINES. Example text to add later: "We also evaluated our framework against external state-of-the-art models such as Model A and

Model B to demonstrate that our baseline neural network matches or exceeds current industry standards for this specific screening task."]

Final Design Adjustment

During testing, we discovered a hidden math error [normalization bug] where the model was learning from raw, unscaled numbers. Fixing this error required clearing the stored data [cache invalidation] and raised the model's test score [validation F1] from 0.4751 to 0.8987.

We removed the step that summarized the entire grid into one single score [global pooling]. We measured that 96 to 99 percent of grid moments have a mix of safe and dangerous wires [strictly mixed frames]. Summarizing the grid erased this critical detail, so we adjusted the model to make a separate prediction for every single wire [edge-level readout]. We also proved that allowing the grid points to talk to each other [message passing] is vital; when we turned this communication off [ablation], the score dropped significantly, proving the map's shape [graph structure] holds important information.

We designed the safety gate to work in one direction only [asymmetric gate]. It will block the model from saying the grid is safe, but it never blocks the model from issuing a warning, because a false alarm is much less costly than a blackout. Finally, we added mandatory tracking for why the artificial intelligence rejects rules. During one test, the artificial intelligence rejected 31 out of 32 rules. When we looked closely, we found the artificial intelligence was entirely inventing fake reasons [fabricated reasoning], such as claiming a word was undefined even though it was given in the instructions.

5.3 Statistical Analysis

To keep our tests strictly fair, we locked our passing grade [held threshold] at 0.8849 based solely on the training map, and we applied this exact same number to all three maps. We avoided picking the best boundary for each individual map [oracle threshold] because that requires looking at the answer key, which is impossible in the real world.

We discovered that the number of examples the model looks at during testing [eval batch size] artificially changes the final scores. Because the model updates its internal math based on the current group of data [live batch statistics], changing the group size from 64 to 512 caused the model's score to shift by 0.028. To keep everything mathematically steady, we pinned the testing group size strictly at 64.

Finally, we measured the exact cost of trying to catch more errors [F-beta trade curve]. The normal passing grade treats missed problems and false alarms equally [F1 objective]. When we adjusted the system to prioritize catching real problems [F2 objective], we found that cutting the missed problems in half on the training grid costs 3.2 extra false alarms for every real problem caught.

Map [Topology]	Goal [Objective]	Passing Grade [Threshold]	Score [F1]	Found Disasters [Recall]	Missed Disasters [Missed]	False Alarms
Familiar [NeurIPS 2020]	Balanced [F1]	0.8849	0.8956	0.902	2,041	2,331
Familiar [NeurIPS 2020]	Catch More [F2]	-0.7896	0.8558	0.953	987	5,689
Large Unfamiliar [WCCI 2022]	Balanced [F1]	0.8849	0.5577	0.629	68,166	115,283
Large Unfamiliar [WCCI 2022]	Catch More [F2]	-0.7896	0.5361	0.689	57,132	162,165

[PLACEHOLDER: STANDARD DEVIATIONS ACROSS MULTIPLE SEEDS.
Example text to add later: "To verify the stability of the neural network,

training was repeated across 3 random seeds. The model achieved a score of 0.8956 ± 0.005 , confirming that the high performance on the training grid is a stable property of the architecture rather than a product of lucky initialization."

5.4 Comparisons and Relationships

There is a clear inverse relationship between the artificial intelligence and the safety gate. As the artificial intelligence's accuracy collapses on new grid layouts, the safety gate's accuracy stays perfectly flat at roughly 93 percent [topology-invariant precision]. Because the artificial intelligence fails much more frequently on the large 118-bus grid, the safety gate is forced to step in and speak much more often [reach], jumping from 0.31 percent of cases on the training grid to 3.04 percent of cases on the large unseen grid.

We also found a surprising relationship regarding grid size: asking the model to move to a map three times larger [scaling up] caused much less performance loss than asking it to move to a smaller 14-bus map [scaling down].

There is a hard structural relationship limiting the safety gate's power. Between 69 and 99 percent of the model's dangerous mistakes happen on grids that look perfectly healthy before the wire is actually removed. Because the safety rules only look at the current picture [present state], they are completely blind to these future failures [structural ceiling]. Adding more rules will never fix this limit, because calculating the future requires a full math simulation [post-contingency power flow].

Map [Topology]	Total AI Mistakes [Dangerous Errors]	Mistakes a Rule Can See [Reachable]	Hard Limit [Structural Ceiling]
Familiar [NeurIPS 2020]	2,041	331	16.2%

Small Unfamiliar [case14]	18,592	95	0.51%
Large Unfamiliar [WCCI 2022]	68,166	21,065	30.9%

5.6 Discussion

The results prove that an artificial intelligence model gets confused in new environments and is not reliable enough on its own for critical safety tasks. A safety layer built from published standards is highly valuable because its accuracy is tied to unchanging physical laws rather than learned guesses.

To prove our final safety rule [thermal constraint] is highly trusted, we built a tracking map [knowledge graph] that proves this exact rule is written in 10 different sections across 4 official documents, published by 2 different international rule-making groups [provenance]. We also measured the actual physics of this rule: if a grid is already overloaded, there is a 91 to 96 percent chance that losing another wire will cause a disaster. Because this number is not exactly 100 percent, it proves the rule provides genuine physical safety guidance rather than just repeating the obvious [label restatement].

We proved that throwing 28 poorly performing rules out of our rulebook actually improved the system's performance, showing that a rulebook full of bad rules completely hides the good ones. Despite the strict limits of what the rules can see, the final surviving rule enforces a vital engineering truth: a grid that is already overloaded cannot possibly be declared safe against further damage.

[PLACEHOLDER: ZERO-SHOT CAPABILITY RETENTION ON EXTERNAL BENCHMARKS. Example text to add later: "To ensure the addition of the symbolic shield did not cause catastrophic forgetting or degrade the model's core processing abilities, we evaluated the shielded model against external diagnostic suites such as [Benchmark A]. The utility drift was minimal ($\pm 0.5\%$), proving the intervention is highly localized and does not damage general competence."]